

Sprint 47: GLM Model Exploration - Abdollah Ghazvanchahi

Feasibility Assessment for EN↔YUE / EN↔CMN Translation

Executive Summary

This report focuses on Sprint 47's evaluation of whether the **GLM model family** (ChatGLM → GLM-4 → GLM-4.5/4.6/4.7 → GLM-5/5.1 from Zhipu AI / Z.ai) can be added to or replace **Qwen2.5** in the **EN↔YUE and EN↔CMN translation pipeline**. The report covers four main areas: (1) research on GLM's training data, tokenizer design, and publicly available information about Cantonese support; (2) a zero-shot evaluation plan using **FLORES-200**, **COMET-22**, and **chrF** metrics; (3) a qualitative evaluation method to identify true Cantonese output versus Traditional Chinese Mandarin output; and (4) a decision framework to determine whether GLM should be adopted, replace an existing model, or be excluded.

Based on the research completed so far, no GLM version up to GLM-5.1 publicly reports dedicated Cantonese training data, a Cantonese-specific tokenizer, or support for Cantonese features such as Jyutping or HKSCS characters. Existing benchmarks, including HKCanto-Eval, HKMMLU, and CantoNLU, show that general-purpose Chinese language models often produce Traditional Chinese text with Mandarin grammar instead of natural Cantonese. HKMMLU also reported that all tested open-source models scored below 75% accuracy on Hong Kong and Cantonese-related tasks.

GLM-4-9B delivers competitive EN↔CMN translation quality, nearly matching the current best model (Qwen3-4B). For EN↔YUE, it performs better than LLaMA 3.1-8B, DeepSeek-R1, and Mistral-7B, but remains behind Qwen3-4B on COMET. The model is therefore a viable candidate for EN↔CMN and shows promising (but not state-of-the-art) results for Cantonese.

1. GLM Model Family Research

1.1 Lineage and Architecture

Zhipu AI (renamed **Z.ai** for international markets and listed on the Hong Kong Stock Exchange in January 2026) has released new GLM models approximately every few months since **GLM-4** was introduced in mid-2024. Among Chinese open-weight model developers, Zhipu has maintained one of the most consistent release schedules.

Model	Release	Scale	Notes
GLM-130B	Aug 2022	130B (dense)	The first major GLM model. It introduced the prefix-LM architecture, using bidirectional attention for input text and causal attention for generated text.
ChatGLM-6B / 2 / 3	2023	6B (dense)	Dialogue-focused models designed for practical use, with a strong focus on Chinese and English.
GLM-4 (9B / 32B)	Jun 2024	9B–32B (dense)	Trained on about 10T tokens with up to 128K context length. Multimodal (GLM-4V) and voice (GLM-4-Voice) versions were released during the same period.
GLM-4.5 / 4.5-Air	Jul 2025	355B / 106B (MoE)	Trained on 22T tokens (15T text + 7T code/reasoning). Introduced large-scale mixture-of-experts (MoE) architecture and reasoning mode support.
GLM-4.5V / 4.6 / 4.6V	Aug–Sep 2025	MoE, vision variants	Added stronger vision capabilities and tool-use support, including optimized inference on some specialized hardware.
GLM-4.7	Dec 2025	MoE	Focused on agentic coding tasks and improved reasoning workflows.
GLM-5 / 5.1	Feb–Apr 2026	744B / 40B (MoE)	Latest flagship models with open weights.

Sources: Zhipu/Z.ai GLM lineage tracker ([presenc.ai](#), May 2026); "ChatGLM: A Family of Large Language Models from GLM-130B to GLM-4 All Tools" ([arXiv:2406.12793](#)); [zai-org/GLM-4 GitHub README](#); [DeepLearning.AI coverage of GLM-4.5](#); [VentureBeat coverage of GLM-4.6V](#); [NVIDIA NIM model card for GLM-4.7](#).

Architecturally, **GLM-130B** introduced a unique **prefix language model (prefix-LM)** approach. It allowed the model to read the input context using bidirectional attention while generating text using causal attention. Later versions, starting with **GLM-4**, moved closer to the standard decoder-only transformer design used by many modern LLMs, while keeping the GLM name and tokenizer foundation.

Starting with **GLM-4.5**, Zhipu adopted a **mixture-of-experts (MoE)** architecture to scale model size more efficiently. Instead of activating the entire model for every request, MoE models activate only selected expert networks, allowing larger total parameter counts while reducing the computation required for each inference step.

1.2 Chinese Pretraining Data

GLM-4-9B was trained on about **10 trillion tokens**. The training data was mainly in Chinese and English, with a smaller amount of data from 24 other languages. The model's alignment and optimization focused mostly on Chinese and English tasks.

GLM-4.5 / 4.5-Air used a larger training dataset of about **22 trillion tokens**, including **15 trillion general text tokens** and **7 trillion tokens focused on code and reasoning tasks**.

Across the four ChatGLM/GLM technical reports reviewed, Zhipu describes common data-processing steps such as removing duplicate data, filtering low-quality content, and cleaning offensive language, placeholder text, and repeated templates. The reports also mention giving more weight to high-quality sources such as books and Wikipedia. However, none of the reports specifically mention using Cantonese, Hong Kong, or Yue-language datasets during training.

Sources: arXiv:2406.12793 (ChatGLM technical report); DeepLearning.AI / The Batch coverage of GLM-4.5 pretraining.

1.3 Tokenizer Design

GLM's tokenizer is based on byte-level BPE. It combines Chinese and multilingual token sets with OpenAI's **cl100k_base (tiktoken)** vocabulary to create a single vocabulary containing 150,000 tokens. This approach is mainly designed to provide strong support for Simplified Chinese, Mandarin, English, and other widely used languages. However, the published tokenizer design does not show any specific focus on Cantonese-related features, such as Hong Kong Supplementary Character Set (HKSCS) characters or Jyutping romanization.

To understand what limited Cantonese support can look like at the tokenizer level, the **CantoNLU** benchmark team created a dedicated Cantonese tokenizer with around 32K tokens, including 27K multi-character tokens. Even this specialized tokenizer covered

only 233 of the 603 HKSCS characters. This shows that Cantonese-specific language resources are still limited across the Chinese NLP field, and models like GLM are likely affected by the same resource limitations.

Sources: [arXiv:2406.12793](#); "CantoNLU: A benchmark for Cantonese natural language understanding" ([arXiv:2510.20670](#)).

2. Cantonese Coverage Assessment

None of the GLM technical reports, from ChatGLM-6B to GLM-5.1, explicitly state that the models were trained on dedicated Cantonese data, fine-tuned for Cantonese instructions, or designed to support Cantonese-specific features such as HKSCS characters or Jyutping tokenization. In contrast, the GLM family clearly advertises support for 24 multilingual languages.

Direct zero-shot evaluation on FLORES-200 provides mixed but informative evidence. While GLM-4-9B achieves competitive COMET and chrF scores on EN→YUE and YUE→EN (COMET 0.8431 / 0.8553, chrF 24.87 / 56.46), these scores remain below the current best model (Qwen3-4B). More importantly, the qualitative analysis of English-to-Cantonese translations shows that GLM-4-9B successfully generates genuine Cantonese using characteristic particles and grammar (e.g., 嘅, 咗, 畀, 喺, 唔, 冇) rather than producing Mandarin written in Traditional Chinese script.

Overall, the combination of public documentation and empirical results indicates that Cantonese was not a major focus of the GLM training and development process. GLM-4-9B can produce recognizably Cantonese output, but its performance is still inferior to Qwen3-4B and does not demonstrate the depth of Cantonese support seen in models explicitly optimized for the language.

3. Candidate Model Selection

Based on the team's available TPU/GPU resources on Google Colab or Kaggle, and the model sizes evaluated during Sprint 44 and Sprint 45 (Qwen3-4B, DeepSeek-R1-0528-Qwen3-8B, LLaMA-3.1-8B, and Mistral-7B), **GLM-4-9B** is the recommended model for evaluation. It is a dense 9B-parameter model with an Apache 2.0 license, supports a 128K context window, and has a size similar to the models already tested. **GLM-4.5-Air** (106B MoE with about 12B active parameters) can be

considered as an alternative only if enough TPU memory is available. However, it has less native support for PyTorch/XLA on TPUs than the dense GLM-4 models, so using it may require more engineering effort compared to the team's previous work with MADLAD-400 and Qwen3 on TPUs.

4. Zero-Shot Evaluation Methodology

4.1 Dataset and directions

The evaluation uses the **FLORES-200 devtest split** with four translation directions: **EN→YUE, YUE→EN, EN→CMN, and CMN→EN**. This follows the same evaluation setup used for **Qwen3, DeepSeek, LLaMA, and Mistral** in Sprint 44/45, allowing direct comparison between models.

4.2 Evaluation Metrics

- **COMET-22 (Unbabel/wmt22-comet-da)**: This is the main evaluation metric and follows the team's standard for ranking translation quality.
- **chrF (SacreBLEU implementation)**: This metric is included because it provides additional insight, especially when COMET and character-level quality differ for Chinese and Cantonese outputs.
- **BLEU and ROUGE**: These metrics are reported only as secondary reference scores, following the same reporting approach used in Sprint 44/45.

4.3 Prompting protocol

The evaluation uses a **zero-shot, single-turn prompt** for each source sentence. No examples are provided, and no additional system instructions are used except specifying the target language and writing style. For example: *"Translate the following English sentence into Cantonese (Hong Kong, Traditional script)."*

This setup measures the model's default translation behavior rather than how well it can be guided with additional instructions. If the model produces Traditional-script Mandarin instead of natural Cantonese, improving its ability to follow Cantonese-specific instructions would be considered a separate future investigation.

4.4 Execution

A ready-to-run Colab/Kaggle notebook, `glm-flores200-zeroshot-eval.ipynb`, is provided with this report. The notebook includes the full evaluation pipeline: loading the GLM model, loading FLORES-200 data, generating translations, calculating COMET-22 and chrF scores, and performing the qualitative Cantonese output analysis described later.

The notebook follows the team’s existing PyTorch/XLA TPU-based workflow. Zero-shot evaluation on FLORES-200 across all four translation directions (EN↔CMN and EN↔YUE) has now been completed using the unsloth/GLM-4-9B-0414-unsloth-bnb-4bit model. Results are presented in Sections 5 and 6 below.

5. Zero-Shot FLORES-200 Evaluation Results – GLM-4-9B

GLM4_9B Results (*unsloth/GLM-4-9B-0414-unsloth-bnb-4bit*):

Direction	COMET	BLEU	chrF	ROUGE-1	ROUGE-2	ROUGE-L
EN→CMN	0.8739	39.91	34.41	0.2585	0.0888	0.2557
CMN→EN	0.8697	27.67	58.41	0.6539	0.3797	0.5749
YUE→EN	0.8553	26.10	56.46	0.6314	0.3605	0.5517
EN→YUE	0.8431	27.35	24.87	0.1894	0.0579	0.1857

Comparison with Previous Models:

- vs. Qwen3-4B (strongest baseline):
GLM-4-9B is slightly behind on EN→CMN (COMET 0.8739 vs 0.8864; BLEU 39.91 vs 41.57) and comparable on CMN→EN and YUE→EN. On EN→YUE, GLM shows similar chrF but slightly lower COMET.
- vs. DeepSeek-R1: GLM-4-9B outperforms on all directions, especially EN→YUE and YUE→EN.
- vs. LLaMA 3.1-8B: GLM-4-9B is clearly stronger across the board.

- vs. Mistral-7B: GLM-4-9B significantly outperforms Mistral on every metric and direction.

Summary of Quantitative Performance:

GLM-4-9B delivers competitive EN↔CMN translation quality, nearly matching the current best model (Qwen3-4B). For EN↔YUE, it performs better than LLaMA 3.1-8B, DeepSeek-R1, and Mistral-7B, but remains behind Qwen3-4B on COMET. The model is therefore a viable candidate for EN↔CMN and shows promising (but not state-of-the-art) results for Cantonese.

6. Recommendation

Recommendation: Adopt GLM-4-9B alongside Qwen3-4B (hybrid pipeline).

Evidence:

COMET Comparison Table (All Models):

Direction	Qwen3-4B	GLM-4-9B	DeepSeek-R1	LLaMA 3.1-8B	Mistral-7B
EN→CMN	0.8864	0.8739	0.8713	0.8455	0.6857
CMN→EN	0.8723	0.8697	0.8602	0.8608	0.8315
YUE→EN	0.8610	0.8553	0.8468	0.8495	0.7934
EN→YUE	0.8642	0.8431	0.8346	0.8085	0.6115

Qwen3-4B remains the strongest model overall on COMET. GLM-4-9B ranks second across all directions and is within ~0.013–0.021 points of Qwen on the three strongest directions. It significantly outperforms LLaMA 3.1-8B, DeepSeek-R1, and Mistral-7B.

chrF Comparison Table (All Models):

Direction	Qwen3-4B	GLM-4-9B	DeepSeek-R1	LLaMA 3.1-8B	Mistral-7B
EN→CMN	35.30	34.41	30.86	28.11	17.32
CMN→EN	57.69	58.41	55.72	55.74	49.50
YUE→EN	56.73	56.46	54.08	55.04	44.81
EN→YUE	24.66	24.87	20.72	20.88	9.92

Key chrF Insights:

- GLM-4-9B achieves the best chrF score on CMN→EN and EN→YUE.
- On EN→CMN and YUE→EN, it is very close to Qwen3-4B (within ~0.9–0.3 points).
- GLM-4-9B substantially outperforms LLaMA, DeepSeek, and Mistral on every direction.

Overall Assessment:

- EN↔CMN: GLM-4-9B is a strong runner-up to Qwen3-4B and a clear improvement over all other tested models.
- EN↔YUE: GLM-4-9B matches or exceeds Qwen3-4B on chrF and remains competitive on COMET, while being markedly better than the other three models.

Decision: Add GLM-4-9B to the translation pipeline as a secondary / complementary model for both language pairs. Continue monitoring GLM-4.5/4.6/4.7 and GLM-5 releases. Do not replace Qwen3-4B at this time.

7. Qualitative Cantonese Output Analysis (EN→YUE)

A sample of 30+ English→Cantonese translations from `GLM4_9B_en_yue_results.csv` was reviewed for genuine Cantonese particles/grammar versus Mandarin-in-Traditional-Chinese output.

Key Observations:

- GLM frequently uses classic Cantonese particles and structures: 嘅, 咗, 畀, 喺, 唔, 冇, 嚟, 呢啲, 個禮拜一, 佢哋, 呢個, 好興奮, 畀人燒毀, etc.

- Many sentences show natural Cantonese word order and colloquial phrasing (e.g., “個禮拜一”, “一嚟細小嘅可印製嘅芯片”, “專家話呢個可能會令到貧窮國家嘅病人可以早啲發現”, “拿度對加拿大嘅頭球記錄為7比2”, “今日嘅最佳選手係...”).
- Some outputs mix in more formal/Mandarin-leaning phrasing or use characters that could appear in both varieties (e.g., occasional “話” instead of more colloquial alternatives, or slightly more written-style constructions).
- Overall, the model produces recognizably Cantonese text rather than pure Mandarin written in Traditional characters. It is noticeably more “Cantonese-flavoured” than Mistral-7B outputs from previous sprints, though still not as consistently natural as the best Qwen outputs.

Conclusion on Qualitative Check: GLM-4-9B successfully generates genuine Cantonese particles and grammar in the majority of sampled sentences. It does not default to Mandarin-in-Traditional-script output. While occasional formal or hybrid phrasing appears, the model demonstrates clear Cantonese capability that warrants inclusion in the pipeline.

8. Conclusion

The Sprint 47 evaluation of the GLM-4-9B model demonstrates that GLM-4-9B is a strong second-tier model for both EN↔CMN and EN↔YUE translation.

Key findings:

- GLM-4-9B delivers competitive-to-strong performance on EN↔CMN, coming within a small margin of the current leader (Qwen3-4B) on both COMET and chrF.
- On EN↔YUE, GLM-4-9B matches or exceeds Qwen3-4B on chrF and remains competitive on COMET while significantly outperforming LLaMA 3.1-8B, DeepSeek-R1, and Mistral-7B.
- Qualitative analysis confirms that GLM-4-9B produces genuine Cantonese output (using characteristic particles such as 嘅, 咗, 畀, 喺, 唔, 冇, etc.) rather than Mandarin written in Traditional Chinese script.

- No evidence was found of dedicated Cantonese training data or tokenizer in GLM releases up to GLM-5.1, consistent with the pattern seen in other general-purpose Chinese models.

Overall recommendation:

Adopt GLM-4-9B alongside Qwen3-4B in a hybrid pipeline for EN↔CMN and EN↔YUE translation. GLM-4-9B is a valuable complementary model that improves robustness and coverage without replacing the current best performer. Future GLM releases (GLM-4.5/4.6/4.7 and GLM-5) should be monitored for potential further gains in Cantonese capability.

A. Notebook and Reproducibility

The complete zero-shot evaluation notebook used in Sprint 47, including the full pipeline for model loading, translation generation, translation extraction, and metric computation, is publicly available on GitHub.

GitHub Repository

<https://github.com/Aghazvan/Vosyn-Supernova-Project/tree/main/Sprint47-GLM-Model-Exploration-Abdollah.Ghazvanchahi>

The repository contains the following files:

- `glm_flores200_zeroshot_eval.ipynb` – Main evaluation notebook
- `README.md` – Detailed usage instructions and final results table
- `results/GLM4_9B_*_results.csv` and `results/GLM4_9B_*_metrics.csv` – Example output files (one per translation direction)